

Recurring Donation — DQ Analysis (Final)

Run: 2026-08-01 · **Rows checked:** 258,465 · **Rules:** 21 total — 14 active gated · 7 controlled-list report-only · **Result:** 11 PASS · 3 FAIL · 0 errors **Affected records:** 54,631 (a record can break more than one rule)

Controlled-list rules — report-only (no assumed values): RD-005, RD-010, RD-011, RD-012, RD-013, RD-014, RD-021. We do **not** assume any value is wrong for these picklist fields; we list the distinct values in §3b and ask stakeholders which are valid.

- Source SQL: [recurring_donation_staging_dq_PROD_framework.sql](#)
- EDA: [EDA_recurring_donation_csv_analysis.md](#)

How to read this: **PASS** = clean, nothing to look at. **FAIL** = the rule found rows that break it (those rows are saved in [dq.dq_exceptions](#)). FAIL is a *review signal*, not always a real defect.

1. Key numbers

Metric	Value
Raw rows	261,577
Distinct IDs / staging rows	258,465
Duplicates removed	3,112
Rules run	21
Active gated rules	14
Controlled-list report-only	7 (RD-005, RD-010–014, RD-021)
Rules failing	3
Records with ≥1 failure	54,631

2. All 21 rules — full status

Rule	Sev	What it checks	Status	Failed
RD-001	CRITICAL	Id is not null	✓ PASS	0
RD-002	CRITICAL	Id is a valid 15/18-char Salesforce ID	✓ PASS	0
RD-004	HIGH	Active donation has amount > 0	✓ PASS	0
RD-005	—	Status — distinct values reported (not gated)	📋 REPORT-ONLY	—

Rule	Sev	What it checks	Status	Failed
RD-006	MEDIUM	Day of month is 1–31 or Last_Day	✓ PASS	0
RD-007	HIGH	Start date ≤ End date	● FAIL	19,864
RD-009	MEDIUM	Campaign exists in <code>raw.salesforce_campaign</code>	✓ PASS	0
RD-010	—	Installment period — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-011	—	Recurring type — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-012	—	Payment method — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-013	—	Donation type — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-014	—	Regional office code — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-015	MEDIUM	Closed donation has a Closed Reason	● FAIL	79,000
RD-016	HIGH	Paid amount is numeric	✓ PASS	0
RD-017	HIGH	Total donation amount is numeric	✓ PASS	0
RD-018	HIGH	Amount is numeric (<i>new</i>)	✓ PASS	0
RD-019	MEDIUM	Installment amount is numeric (<i>new</i>)	✓ PASS	0
RD-020	MEDIUM	Failed-payments count is a non-negative number (<i>new</i>)	✓ PASS	0
RD-021	—	Currency code — distinct values reported (not gated)	📋 REPORT-ONLY	—
RD-022	MEDIUM	Start/End/Next dates are parseable (<i>new</i>)	✓ PASS	0
RD-023	HIGH	Active donation has a Next Payment Date (<i>new</i>)	● FAIL	90

📋 **RD-005, RD-010, RD-011, RD-012, RD-013, RD-014, RD-021** are **report-only** (controlled-list fields — no assumed values; distinct values in §3b), not counted in PASS/FAIL. RD-018–RD-023 were added after the

data investigation: RD-018–RD-022 are clean guards (0 failures), RD-023 found 90 real issues.

2b. Table creation, dedup & staged-row evidence

`staging.recurring_donation_latest` is rebuilt from `raw.salesforce_recurring_donation`, deduped by `Id` (latest `SystemModstamp`), excluding soft-deleted rows → **258,465 rows, 31 columns** (3,112 duplicates collapsed).

Dedup logic:

```
WITH dedup AS (
  SELECT [Id],[IsDeleted],[SystemModstamp],
    ROW_NUMBER() OVER (
      PARTITION BY CONVERT(VARCHAR(18),[Id])
      ORDER BY COALESCE(TRY_CONVERT(DATETIME2(7),[SystemModstamp],127),
        TRY_CONVERT(DATETIME2(7),[SystemModstamp])) DESC) AS rn
  FROM [raw].[salesforce_recurring_donation]
  WHERE [Id] IS NOT NULL
)
SELECT * FROM dedup
WHERE rn=1
  AND COALESCE(LOWER(LTRIM(RTRIM(CONVERT(NVARCHAR(20),[IsDeleted])))),N'false')
  NOT IN (N'true',N'1',N'yes',N'y');
```

Staging columns (31):

#	Column	#	Column
1	row_number	17	npe03__Installment_Period__c
2	Id	18	npsp__Day_of_Month__c
3	IsDeleted	19	npe03__Next_Payment_Date__c
4	Name	20	Donation_Type__c
5	CurrencyIsoCode	21	npsp__PaymentMethod__c
6	npe03__Contact__c	22	Regional_Office_Code__c
7	npe03__Organization__c	23	npe03__Recurring_Donation_Campaign__c
8	npe03__Amount__c	24	Number_of_Failed_Payments__c
9	npe03__Installment_Amount__c	25	SystemModstamp
10	npe03__Paid_Amount__c	26	_etl_source
11	Total_Donation_Amount__c	27	_etl_source_object
12	npsp__Status__c	28	_etl_loaded_at_utc
13	npsp__RecurringType__c	29	staging_is_duplicate

#	Column	#	Column
14	npsp__StartDate__c	30	staging_duplicate_count
15	npsp__EndDate__c	31	staging_created_at
16	npsp__ClosedReason__c		

Staged-row evidence (SSMS-ready):

```
SELECT TOP (2) [Id],[npsp__Status__c],[npe03__Amount__c],[npsp__StartDate__c],
[npsp__EndDate__c],[npe03__Next_Payment_Date__c]
FROM staging.recurring_donation_latest ORDER BY staging_created_at DESC;
```

Id	Status	Amount	StartDate	EndDate	NextPayment
a094J00000d03BWQAY	Closed	40.0	2021-09-01	2025-12-02	NULL
a094J00000Ms41cQAB	Closed	35.0	2020-06-01	2021-10-25	NULL

3. The 3 failing rules — detail and quality impact

Each failure maps to a **data-quality dimension**: **Completeness** (value missing), **Validity** (value wrong / out of range), **Consistency** (values contradict each other), **Integrity** (broken link between records).

● RD-015 · Closed donation with no Closed Reason — 79,000 · *Completeness*

- **What:** donation is **Closed** but **npsp__ClosedReason__c** is blank (42% of all Closed).
- **Quality impact:** churn/retention reporting is **blind** — we can't tell *why* recurring gifts ended (card failure vs donor cancel vs migration). Cripples win-back targeting, lapse forecasting, and “reason for leaving” dashboards. The data isn't wrong, but this is the single biggest completeness gap in the object.
- **Sample:** **a094J00000d011k**

```
-- RD-015: Closed with no Closed Reason (raw + staging, top 10)
SELECT TOP 10 s.[Id],
      s.[npsp__Status__c] AS stg_status, s.[npsp__ClosedReason__c] AS
stg_closed_reason,
      r.[npsp__Status__c] AS raw_status, r.[npsp__ClosedReason__c] AS
raw_closed_reason
FROM staging.recurring_donation_latest s
OUTER APPLY (SELECT TOP 1 r0.[npsp__Status__c], r0.[npsp__ClosedReason__c]
      FROM raw.salesforce_recurring_donation r0
      WHERE LTRIM(RTRIM(r0.[Id])) = LTRIM(RTRIM(s.[Id]))
      ORDER BY TRY_CONVERT(DATETIME2, r0.[SystemModstamp]) DESC) r
WHERE UPPER(LTRIM(RTRIM(COALESCE(s.[npsp__Status__c], '')))) = 'CLOSED'
      AND NULLIF(LTRIM(RTRIM(COALESCE(s.[npsp__ClosedReason__c], ''))), '') IS NULL;
```

Sample rule matches (actual output — top 8):

Id	npsp__Status__c	npsp__ClosedReason__c
a094J00000d01IkQAA	Closed	NULL
a094J00000d02swQAA	Closed	NULL
a094J00000d02WgQAI	Closed	NULL
a094J00000d03BMQAY	Closed	NULL
a094J00000d0645QAA	Closed	NULL
a094J00000d06reQAA	Closed	NULL
a094J00000d07w4QAA	Closed	NULL
a094J00000d07y5QAA	Closed	NULL

● RD-007 · Start date after End date — 19,864 · Consistency

- **What:** `npsp__StartDate__c` is later than `npsp__EndDate__c` on the same row.
- **Quality impact:** every duration / active-period calculation breaks (tenure, lifetime-value windows, “active on date X”). Time-series and cohort analytics silently return wrong numbers for these rows. Most likely a migration artifact, but it corrupts any date-range logic until fixed or excluded.
- **Sample:** `a094J00000Zmsjc` (Start 2021-07-09 / End 2021-07-08)

```
-- RD-007: Start date after End date (raw + staging, top 10)
SELECT TOP 10 s.[Id],
    s.[npsp__StartDate__c] AS stg_start, s.[npsp__EndDate__c] AS stg_end,
    r.[npsp__StartDate__c] AS raw_start, r.[npsp__EndDate__c] AS raw_end
FROM staging.recurring_donation_latest s
OUTER APPLY (SELECT TOP 1 r0.[npsp__StartDate__c], r0.[npsp__EndDate__c]
    FROM raw.salesforce_recurring_donation r0
    WHERE LTRIM(RTRIM(r0.[Id])) = LTRIM(RTRIM(s.[Id]))
    ORDER BY TRY_CONVERT(DATETIME2, r0.[SystemModstamp]) DESC) r
WHERE NULLIF(LTRIM(RTRIM(COALESCE(s.[npsp__StartDate__c], ''))), '') IS NOT NULL
    AND NULLIF(LTRIM(RTRIM(COALESCE(s.[npsp__EndDate__c], ''))), '') IS NOT NULL
    AND TRY_CONVERT(DATETIME2, s.[npsp__StartDate__c]) > TRY_CONVERT(DATETIME2, s.
[npsp__EndDate__c]);
```

Sample rule matches (actual output — top 8):

Id	npsp__StartDate__c	npsp__EndDate__c
a094J00000d03Q3QAI	2021-08-01	2021-07-27
a094J00000d06AyQAI	2021-09-01	2021-08-11
a094J00000d0HWpQAM	2021-09-01	2021-07-30
a094J00000d0lkPQAQ	2021-09-01	2021-08-18
a094J00000d0O33QAE	2021-09-01	2021-08-04

Id	npsp_StartDate__c	npsp_EndDate__c
a094J00000d0Y5zQAE	2021-09-01	2021-08-12
a094J00000d14mLQAQ	2021-09-01	2021-08-12
a094J00000Mryd4QAB	2021-03-09	2021-03-04

● RD-023 · Active donation with no Next Payment Date — 90 · *Completeness (new rule)*

- **What:** status is **Active** but **npe03__Next_Payment_Date__c** is blank.
- **Quality impact:** collections and cash-flow forecasting **miss these gifts** — an active recurring donation with no scheduled next charge won't be billed or projected. Low count but high operational value: this is direct **lost-revenue risk**, not just a reporting gap.
- **Sample:** a094J00000UJmhq

```
-- RD-023: Active donation with no Next Payment Date (raw + staging, top 10)
SELECT TOP 10 s.[Id],
             s.[npsp__Status__c] AS stg_status, s.[npe03__Next_Payment_Date__c] AS
stg_next_payment,
             r.[npsp__Status__c] AS raw_status, r.[npe03__Next_Payment_Date__c] AS
raw_next_payment
FROM staging.recurring_donation_latest s
OUTER APPLY (SELECT TOP 1 r0.[npsp__Status__c], r0.[npe03__Next_Payment_Date__c]
             FROM raw.salesforce_recurring_donation r0
             WHERE LTRIM(RTRIM(r0.[Id])) = LTRIM(RTRIM(s.[Id]))
             ORDER BY TRY_CONVERT(DATETIME2, r0.[SystemModstamp]) DESC) r
WHERE UPPER(LTRIM(RTRIM(COALESCE(s.[npsp__Status__c], '')))) = 'ACTIVE'
      AND NULLIF(LTRIM(RTRIM(COALESCE(s.[npe03__Next_Payment_Date__c], ''))), '') IS
NULL;
```

Sample rule matches (actual output — top 8):

Id	npsp_Status__c	npe03__Next_Payment_Date__c
a094J00000UJmhqQAD	Active	NULL
a094J00000UJnBGQA1	Active	NULL
a094J00000UJpDCQA1	Active	NULL
a094J00000UJpeNQAT	Active	NULL
a094J00000UJvAKQA1	Active	NULL
a094J00000UJwJxQAL	Active	NULL
a094J00000UJxV7QAL	Active	NULL
a094J00000UJxxXQAT	Active	NULL

Worst records (concrete examples for stakeholders)

Among the active rules, the worst records break **2 rules** — **RD-007 + RD-015** (bad dates + no closed reason), e.g. `a094J00000d0Y5z`, `a094J00000MryeX`, `a094J00000MryYo`. Use one of these as the “look how bad a single record can be” example.

Net quality read: the object is **structurally sound** (all identity, format, and numeric rules pass). The real issues are **completeness** (RD-015, RD-023) and **consistency** (RD-007) — mostly governance and migration cleanup, plus one operational revenue risk (RD-023). Controlled-list fields (Status, Payment Method, etc.) are **not gated** — their distinct values are listed in §3b for stakeholders to confirm.

3b. Controlled-list fields — distinct values for stakeholder review

We do **not** assume any of these values is wrong. The rules for these fields (RD-005, RD-010, RD-011, RD-012, RD-013, RD-014, RD-021) are **report-only** (not gated). Below are the actual distinct values and counts — please confirm which are valid / canonical (and whether near-duplicates like `SEPA` vs `sepa_debit` should be merged).

Field	Distinct values (count)
<code>npsp__Status__c</code>	Closed 188,618 · Active 69,817 · Paused 27 · Lapsed 2 · <code><NULL/BLANK></code> 1
<code>npe03__Installment_Period__c</code>	Monthly 180,606 · <code><NULL/BLANK></code> 45,487 · Daily 16,351 · Yearly 13,004 · Weekly 3,017
<code>npsp__RecurringType__c</code>	Open 203,915 · Fixed 54,548 · <code><NULL/BLANK></code> 2
<code>npsp__PaymentMethod__c</code>	card 97,802 · Direct Debit 66,863 · Card Payment 52,734 · <code><NULL/BLANK></code> 29,812 · <code>sepa_debit</code> 10,778 · <code>us_bank_account</code> 232 · <code>SEPA</code> 129 · <code>bacs_debit</code> 34 · Cash 19 · <code>ach_debit</code> 17 · <code>acss_debit</code> 16 · Check 14 · Bank Transfer 6 · ACH 4 · Cheque 2 · Credit Card 1 · Card Terminal 1 · BankTransfer 1
<code>Donation_Type__c</code>	RD 177,586 · 10N 37,357 · 30N 13,433 · RDA 12,887 · 10D 11,048 · EMI 3,136 · TJC 3,017 · <code><NULL/BLANK></code> 1
<code>Regional_Office_Code__c</code>	UK 134,715 · FR 88,007 · US 20,467 · AR 7,437 · CA 3,283 · ES 2,146 · BE 1,351 · IE 1,002 · WQ 51 · <code><NULL/BLANK></code> 6
<code>CurrencyIsoCode</code>	GBP 133,633 · EUR 93,488 · USD 27,211 · CAD 3,280 · AED 648 · SAR 139 · QAR 66

```
-- Re-run the distinct list for any controlled-list field (change the column)
SELECT COALESCE(NULLIF(LTRIM(RTRIM([npsp__PaymentMethod__c])), ''), '<NULL/BLANK>')
AS value, COUNT(*) AS cnt
FROM staging.recurring_donation_latest
GROUP BY
COALESCE(NULLIF(LTRIM(RTRIM([npsp__PaymentMethod__c])), ''), '<NULL/BLANK>')
ORDER BY cnt DESC;
```

3c. Rule match summary (ranked)

Authoritative failed counts (framework run, `dq.dq_results`): 98,954 matches across **54,631 distinct records** (one record can break more than one rule). Only the 3 active gated rules below failed; the other 11 active gated rules PASS with 0.

By severity (active failing rules)

Severity	Matches	% of fail total
CRITICAL	0	0.0%
HIGH	19,954	20.2%
MEDIUM	79,000	79.8%
LOW	0	0.0%
Total	98,954	100.0%

By rule (ranked)

Rank	Rule	Description	Matches	Severity
1	RD-015	Closed donation with no Closed Reason	79,000	MEDIUM
2	RD-007	Start date after End date	19,864	HIGH
3	RD-023	Active donation with no Next Payment Date	90	HIGH
Total			98,954	

`dq.dq_exceptions` stores a **capped subset** of these rows (RD-015 capped at 50,000) — see §4c. The counts above (from `dq.dq_results`) are the authoritative run results.

4. Business Actions Required

Rule	Count	Stakeholder question	Decision options	Owner	Deadline
RD-015	79,000	Is Closed Reason mandatory for Closed donations?	Enforce at source; make optional (drop rule); backfill from history	RD Business Owner	This week
RD-007	19,864	Are Start > End dates a migration artifact or real errors?	Fix at source; accept as legacy; exclude from date logic	RD Business + Data Eng	This week
RD-023	90	Should an Active donation always have a Next Payment Date ?	Require it; allow blank; auto-derive next date	Collections Owner	This week
RD-005/010–	report-only	Which controlled-list values are canonical	Confirm/curate allowed list; merge near-	RD Business Owner	Next week

Rule	Count	Stakeholder question	Decision options	Owner	Deadline
014/021		(§3b)? e.g. is SEPA = sepa_debit ?	duplicates		

Open questions found in the data (not yet rules — need a decision)

- **Total_Donation_Amount__c meaning:** it is **not** the lifetime total. Evidence shows Paid (e.g. 2,125) far exceeds Total (35); **Total_Donation_Amount__c** tracks the **recurring/installment amount**, and **npe03__Paid_Amount__c** is the lifetime paid. Please confirm the intended meaning before any "paid vs total" rule.
- **At-risk donors:** 3,286 Active donations have **≥3 failed payments** — do you want an alert/monitoring rule and at what threshold?
- **Open vs Fixed lifecycle:** 97,271 "Open" have an End date and 24,151 "Fixed" have none — confirm the expected rule before enforcing.

4b. Most-violating records

Ranked over the **active** failing rules (RD-007 / RD-015 / RD-023):

```
SELECT TOP 5 e.record_id, COUNT(DISTINCT r.check_name) AS rules_failed,
      STRING_AGG(r.check_name, ', ') WITHIN GROUP (ORDER BY r.check_name) AS
failed_rules
FROM dq.dq_exceptions e JOIN dq.dq_rule_catalog r ON r.rule_id=e.rule_id
WHERE r.object_name='Recurring_Donation' AND e.resolution_status='OPEN'
      AND r.check_name IN ('RD-007','RD-015','RD-023')
GROUP BY e.record_id ORDER BY rules_failed DESC;
```

Actual output (top 5):

record_id	rules_failed	failed_rules
a094J00000d0Y5zQAE	2	RD-007, RD-015
a094J00000MryeXQAR	2	RD-007, RD-015
a094J00000MryYoQAJ	2	RD-007, RD-015
a094J00000MryZAQAZ	2	RD-007, RD-015
a094J00000MrzhIQAR	2	RD-007, RD-015

Worst case = a closed donation with **inverted dates AND no closed reason** — a consistency *and* a completeness defect on the same record.

4c. What's stored where + framework status

- ☒ Registered in **dq.dq_rule_catalog** — 21 active RD rules. Framework run **2026-08-01**.
- ☒ 14 active gated · 7 report-only (RD-005/010–014/021).

-  **dq.dq_exceptions** holds **69,954 OPEN** rows for the active failing rules, not the authoritative **98,954** — the runner **capped RD-015 at 50,000** (**MaxExceptionsPerRule**). Authoritative failed counts live in **dq.dq_results**. **Cleanup task:** re-run RD-015 uncapped if full exception rows are needed.
-  Business review pending — exceptions stay **OPEN** until reviewed.
-  No writeback to Salesforce (not enabled).

Table	Rows	Status
staging.recurring_donation_latest	258,465	latest-per-Id, 31 columns
dq.dq_exceptions (RD, OPEN)	69,954	RD-015 capped at 50k (from 79k)
dq.dq_rule_catalog (RD)	21	14 gated · 7 report-only

4d. Sign-off & next meeting

What we accomplished

- Staged 258,465 donations (3,112 duplicates collapsed).
- Ran 21 rules (14 gated · 7 report-only); 3 failing, 0 CRITICAL.
- 98,954 matches across 54,631 records; identified 4 decisions + 1 exceptions-cleanup task.

Decisions needed

- ☐ RD-015 Closed Reason mandatory? _____
- ☐ RD-007 Start>End fix-at-source vs accept? _____
- ☐ RD-023 Active must have Next Payment Date? _____
- ☐ Controlled-list canonical values (§3b)? _____
- ☐ Approve exceptions cleanup (uncap RD-015)? _____